State Recognition of Electric Control Cabinet Switches Based on CNNs

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***Abstract*—When using the CNNs for image processing, it is more easy to learn the invariant features of the image without too much preprocessing. This is important for the task of image classification and recognition which need to extract deep features of the object. This paper proposes a switch state recognition algorithm which combines digital image processing technique and CNNs for the electrical control cabinet switches in substations. Firstly, use perspective transformation method to rectify the original image which is positioned and segmented using the projections of horizontal and vertical gradient. Then label the segmented images to get sample sets for training CNNs later. After that, design the architecture of CNNs with image processing knowledge and use the designed CNNs to extract the key features of switch state, build and train the CNNs model using labeled "disconnected" and "closed" switches. Finally, the state of the switch is recognized by the trained model and showed in the image. The experimental results show that the algorithm has the characteristics of high recognition accuracy, good robustness and fast recognition speed**.

Keywords-CNNs; state of switch; segmenting; recognition

I. INTRODUCTION

With the scale of domestic electric power network continuously expand, the relationship between the power equipment is becoming more and more complex. The number of switch control cabinet in substation is growing, and it's essential to get the state of switches in real time for substation is an important part of the power grid, the operation state of substation is one of the key factors to determine whether the power grid can operate safely [1]. However, current manual inspection is time and labor consuming and it is easy to make error or leak during detecting due to environment and exhaustion. Therefore, it is very important to use the software system to automatically recognize the status of the switch, which is also an important link in the unattended development trend of substation [2]. It has been widely used in the electrical industry to control and protect the power system by means of image processing technology auxiliary automation equipment recently years [3]. For example: the algorithm of classification and recognition based on traditional projection method, a template matching algorithm based on image shape feature analysis [4], the algorithm based on Hough Forest, an algorithm based on SURF which using GPRS to transmit the switch monitoring image and the fault information to the control room for precise matching and verification, etc. These methods can improve the quality of unattended substation and improve the safety supervision efficiency [5]. As a powerful visual model, CNNs have demonstrated remarkable performance in various visual recognition problems, and attracted considerable attention in recent years [6]. Using the digital image processing technology and the full convolution neural networks algorithm, the calculation speed is faster and the recognition accuracy is higher [7]. In this paper, a convolutional neural network is designed for the state recognition of switches commonly used in substations. Taking pictures using camera, and then segmenting, labeling the large number of images for training to obtain the model parameter files. Finally, the algorithm is transplanted into Android application, which realizes recognizing the state of switches automatically with the inputting image. And the whole system has been used in the actual inspection of substation.

II. ALGORITHM OVERVIEW

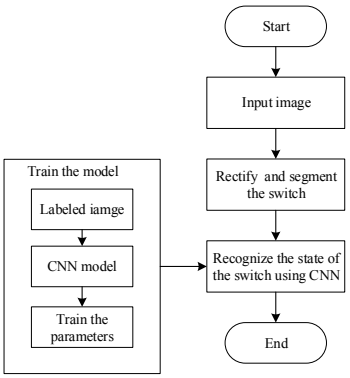


Figure 1. Flow chart of switch status recognition

Figure 1 shows the overall framework diagram of the algorithm in this article. The algorithm contains two parts. The First part is the process of training the CNNs model. The second part is the process of recognizing the state of the switches. The algorithm firstly preprocesses the image to rectify the switch panel and segment every single switch from the panel. Using the segmented switches as training and test sets to train the CNNs model, then using CNNs model to recognize the state of the switch.

III. IMAGE PREPROCESSING

The main work of image preprocessing is rectifying and segmenting the original image to get single switch image. Then the single switch image is sent to CNNs model to be recognized. The whole process is shown in Figure 2.

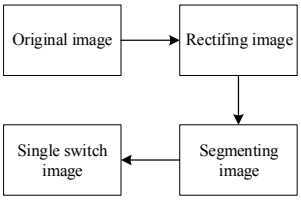


Figure 2. Flow chart of image preprocessing

Using perspective transformation to rectify the original image. Then computing the image gradient to obtain the horizontal and vertical projection. Finally, segmenting the switches via the rows and columns cycle obtained by analyzing the projection. The result is shown in Figure 3.

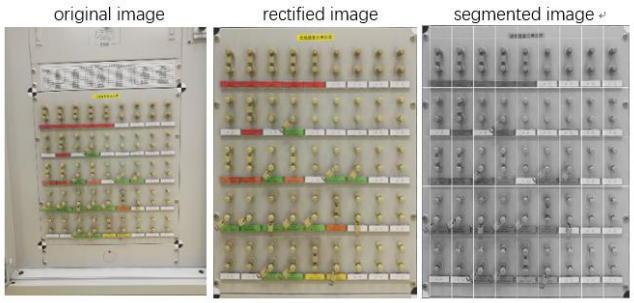


Figure 3. Image preprocessing

IV. TRAINING THE MODEL

The training process of the CNNs model includes network structure design, data set production, and fine tuning of super parameters，each of them interacts with each other. The design of network structure requires rich experience in image processing to design the kernel size and step size of the convolution and pooling layers. And choose the appropriate normalization and activation function. Batch normalization allows us to use much higher learning rates and be less careful about initialization [8]. The production of training data sets should be abundant and reasonable. The super parameters need to be set according to the application scenario. Such as setting the appropriate learning strategy and learning rate and so on. Every of these process has a great impact on the final outcome. The specific process is shown in Figure 4.

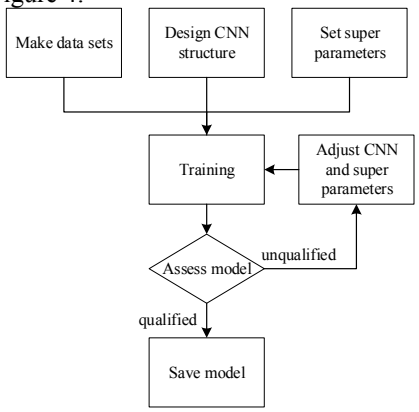


Figure 4. The process of training model

A. *The Architecture of CNNs*

The high complexity of object recognition tasks means that it is difficult to externalize problems even with large data sets. Traditional image processing technology uses a variety of filters to perform image preprocessing and feature extraction. The performance of the algorithm depends entirely on the designer's prior knowledge, to design the nuclear parameters of the filter, and to manually adjust the kernel parameters to achieve the purpose of extracting features. Convolution network is a multilayer perceptron specially designed for identifying the two-dimensional shape, its learning ability can be adjusted by the depth and width of the control network. It can also make powerful and accurate assumptions for high-level attributes of the image, including statistical stability and local features relied on by pix [9]. This network structure is highly invariable to translation, scaling, slant or other forms of deformation. The main advantage of CNNs is that the whole system is end-to-end trained, from the original pixel to the final category, thus reducing the need for manually designing the appropriate feature extractors [10]. After designing the network architecture, use the prepared training set to learn automatically, when training a convolutional layer of the CNNs, it's actually training a series of filters. Making these filter groups have high activation of specific patterns to achieve the classification or detection tasks.

Aiming at the recognition task, combined with certain prior knowledge and classic image classification network to continuously adjust the parameters and conduct training and testing. Finally, the optimal CNNs structure is shown in Figure 5.

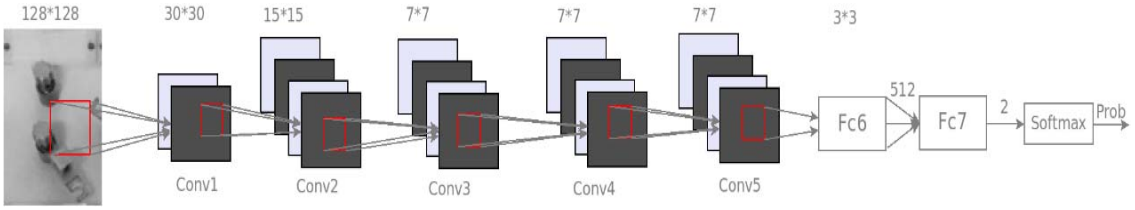


Figure 5. CNNs structure diagram

The size of inputting switch image is resized to 128*128. The probability values in "closed" state and "disconnecting" state are calculated by Softmax function after five convolution layers and two full connection layers. The first five layers of convolution are used to extract key features of the state of the switch, including the pooling layer and activation function to assist convolution layer mapping the original data to the hidden feature space. The full connection layer integrates the depth characteristics of the convolution layer and acts as a "classifier" in the whole network to realize an end-to-end architecture. More attention should be payed to the dropout because it is known to work well in fully-connected layers for deep CNNs [11].

B. Extract Features by Convolution Layer

Convolution is a local operation, which can obtain local information of the image through the convolution kernel with a certain size. A convolutional layer is parametrized by the size and the number of the maps, kernel sizes, skipping factors, and the connection table [12]. And multiple times convolution operation can extract significant features and reduce the feature dimension and computation. After all the convolution operation, multiple 3*3 feature graphs are obtained from the original switch image. The calculation formula of image width and height after convolution is as follows:

$$\begin{cases} \text{output}_w = [\dfrac{image_w + 2\,pad - \ker nel_{size}}{stride}] + 1 \\ \text{output}_h = [\dfrac{image_h + 2\,pad - \ker nel_{size}}{stride}] + 1 \end{cases} \quad (1)$$

The first layer has 48 convolution kernel, the kernel size is 11, the step length is 4, the boundary is not filled, and 48 feature graphs of size 30*30 are obtained after calculation. And second layer has 128 convolution kernel, the kernel size is 5, the step length is 1, the boundary pad is 2, and 128 feature graphs of size 15*15 are obtained after calculation. Both the first and second layers have ReLU activation layer, LRN layer and maximum pooling layer, and the core size of the pool layer is 3, and the step length is 2, without filling.

The third, fourth and fifth layer has the same parameters, the convolution kernel amount is 128, the kernel size is 3, the step length is 1, the boundary pad is 1, and the ReLU activation layer is followed by each layer, and 128 feature graphs of size 7*7 are obtained after calculation. After the maximum pooling of the fifth convolution layer, multiple 3*3 feature graphs are obtained. The convolution layer, pooling layer and activation layer are used to map the original data to the hidden feature space which is good for the state recognition. The full connection layer of the sixth and seventh layers will map the learned features to the marker space of the sample to make predictions. Figure 6 shows the results after first three convolution layers.

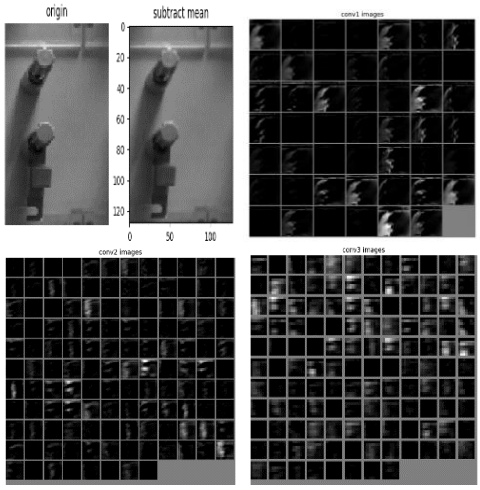


Figure 6. Original iamge and feature graphs after convolution

C. Train and Save the Model.

In this paper, Caffe is used as a deep learning framework for the training of CNNs to train the model. Finally, the model is used to recognize the switch state of electrical control cabinet. The specific steps are as follows:

- Make data sets. Segment switch panel images collected from different angles and light shadows to get single switches. A total of 1600 single switch images were selected as the training set, and 300 images were used as the test set. In order to ensure the balance, the "closed" and "disconnected" state of the switch images each accounted for 50%.
- Set super parameters. In the Solver file, set SGD as optimization algorithm, studying step by step as learning strategies, base learning rate is 0.01, every 200 iterations, learning rate decreases an order of magnitude, and using CPU mode training.
- Train and save the model. The whole neural network is trained by using the labeled switch training set. During the training, the loss value decreased steadily, and the recognition rate gradually increased and finally stabilized at about 96%. The loss value and accuracy curve of the training process are shown in Figure 7 and Figure 8.

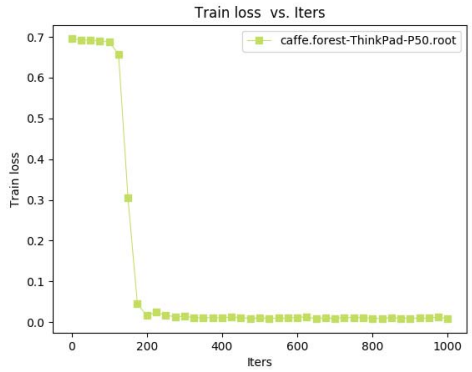


Figure 7. Training loss diagram

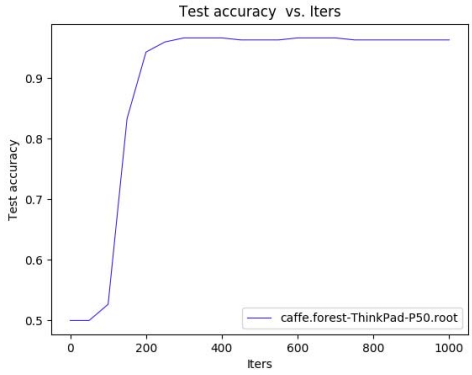


Figure 8. Testing accuracy diagram

As can be seen from the figure, when the number of iterations reaches 400, the accuracy is basically stable and the value of the loss is stable within 0.02. Finally save the model file after 1000 iterations.

V. EXPERIMENTS AND ANALYSIS

Caffe provides multimedia scientists and practitioners with a clean and modifiable framework for state-of-the-art deep learning algorithms and a collection of reference models [13]. Based on the actual problems, this paper uses Caffe framework and Android phone as the test platform to test the CNNs model and the whole system respectively. The model was tested using 300 switch images of the test set, with 150 are "closed" and 150 "disconnected". Using the Caffe interface to load the model and validate the image, and output the results of the model prediction. The experimental results are shown in the table below. The test cell phone used is honor 8, and the time consumed with 5*9 switch panel is about 1.5 seconds, which can meet the real-time application. The traditional algorithm of experimental comparison uses the projection method to extract the gradient features of X and Y direction, and then use the SVM brought by Opencv to train the classifier. Experiments show that the algorithm achieves optimal balance in recognition rate and operation speed, and has robustness under the influence of certain degree of slant and shadow. The actual recognition effect diagram is shown in Figure 9.

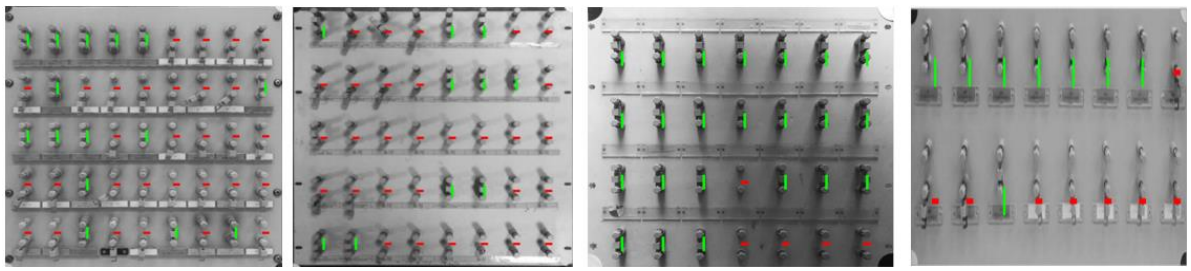


Figure 9. Results of state recognition

TABLE I. CONTRAST TEST RESULTS

The status of slant	*The status of shadow*	*Recognition rate of traditional algorithm*	*Recognition rate of current algorithm*
No	No	86.3	96.6
No	Yes	77.4	91.2
Yes	No	79.5	91.1
Yes	Yes	71.8	88.5

TABLE II. CONTRAST TEST OF DISCONNECTED AND CLOSED STATE

state of the switch	*Recognition rate of traditional algorithm*	*Recognition rate of current algorithm*
disconnected	75.8	94.29
closede	76.4	96.57

As can be seen from the above table, this algorithm has a great improvement over the traditional algorithm in recognition rate, and the algorithm has better performance and robustness in the case of shadow and slant. However, the test results of the switch images with the disconnected and the closed state are quite different. The recognition effect of the closed state is quite good, and the state recognition of the disconnected state is not satisfactory. By observing and analyzing unrecognized switch images, three causes are found as followed:

1) The condition of disconnected switch is changeable and the training sample is not abundant.
2) When the shadow is serious, the shadow is connected to the switch so that the model predicts it is closed.

3) The switch segment is not ideal, resulting in one switch image contains parts of another switch.

The above analysis shows that the image preprocessing has a great influence on the model recognition rate. At the same time, in order to cope with the switch state in different situations, a larger number of training samples are required under different circumstances.

VI. CONCLUSION

Aim at state recognition of the electrical control cabinet switch in transformer substation， using traditional image processing technique to preprocess and segment the switch image. Then using the segmented switch image to train the CNNs model and recognize the state of the switch. The algorithm makes up the defect of traditional image processing recognition and it has high accuracy, good robustness, and can be used in real-time scenarios. After migrating learning and retrained, the model can be used in other scenario. What is more, it is of great practical significance to transfer the algorithm to Android devices for real-time detection and recognition. Due to the image preprocessing has a great influence on the model recognition rate, the algorithm need to find a better algorithm for locating and segmenting the switches.

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REFERENCES

[1] DING Sihai, LIU Yuxue, LU Linji. Application of digital image processing in the switch identification[J]. Microcomputer Application,2013, 30(5): 39-40.

[2] SHAO Jianxiong, YAN Yunfeng, QI Donglian. Substation switch detection and state recognition based on Hough forests[J]. Automation of Electric Power Systems, 2016, 40(11): 115-120.

[3] ZHANG Yilian,HUANG Lilei,QIAN Zhong, et al. Video deshaking based status detection of indoor switch cabinets.

[4] ZHANG Hao, WANG Wei, XU Lijie, et al. Application of image recognition technology in electrical equipment on-line monitoring[J]. Power System Protection and Control, 2010, 6: 88-91.

[5] GONG Xin, MENG Lingfeng, ZHANG tao, et al.Monitoring and identification system of subatation switch division based on SURF[J].Instrument technique and sensor,2017(8):56-58.

[6] Yu S, Jia S, Xu C. Convolutional neural networks for hyperspectral image classification[J]. Neurocomputing, 2017, 219: 88-98.

[7] CHEN Tuo. Switch state recognition method of intelligent substation based on full convolution neural network[A]. Proceedings of the 35th Chinese Control Conference(F)[C].2016:4.

[8] Ioffe S, Szegedy C. Batch normalization: Accelerating deep network training by reducing internal covariate shift[J]. arXiv preprint arXiv:1502.03167, 2015.

[9] Krizhevsky A, Sutskever I, Hinton G E. Imagenet classification with deep convolutional neural networks[C]//Advances in neural information processing systems. 2012: 1097-1105.

[10] Sermanet P, Eigen D, Zhang X, et al. Overfeat: Integrated recognition, localization and detection using convolutional networks[J]. arXiv preprint arXiv:1312.6229, 2013.

[11] Wu H, Gu X. Towards dropout training for convolutional neural networks[J]. Neural Networks, 2015, 71: 1-10.

[12] Ciresan D C, Meier U, Masci J, et al. Flexible, high performance convolutional neural networks for image classification[C]//IJCAI Proceedings-International Joint Conference on Artificial Intelligence. 2011, 22(1): 1237.

[13] Jia Y, Shelhamer E, Donahue J, et al. Caffe: Convolutional architecture for fast feature embedding[C]//Proceedings of the 22nd ACM international conference on Multimedia. ACM, 2014: 675-678.